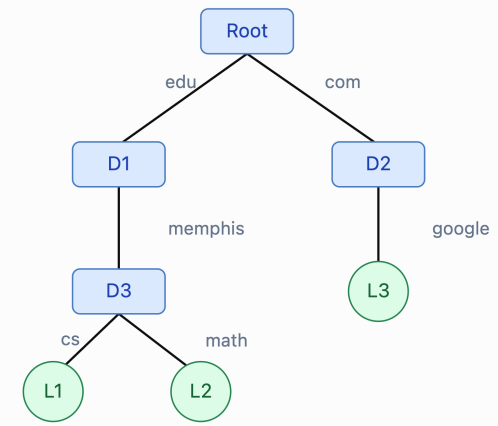
Exercise 10
Name:_____

Consider a distributed file system in which resources are organized into a hierarchical name space. The system contains the following directory nodes and leaf nodes.



Assume that leaf nodes contain the following addresses:

L1 192.168.1.10

L2 192.168.1.20

L3 192.168.2.10

1. Construct the directory table for each directory node (Root, D1, D2, and D3). Each directory table should contain pairs of edge labels and target node identifiers. (Write next to each directory node in the figure above).

Root Directory Table:
- (edu, D1)
- (com, D2)

D1 Directory Table:
- (memphis, D3)

D2 Directory Table:
- (google, L3)

D3 Directory Table:
- (cs, L1)
- (math, L2)

2. Write the absolute path name for each of the three leaf nodes, starting from the root node. Use the notation Root:[label1, label2, ...]. (Write next to each leaf node).

L1: Root:[edu, memphis, cs]
L2: Root:[edu, memphis, math]
L3: Root:[com, google]

3. Assume that Root, D1, and D3 are managed by three different name servers, S0, S1, and S2, respectively. Describe how the client resolves Root:[edu, memphis, cs] using iterative name resolution. Specify the request and response messages exchanged between the client and each name server.

Step 1: Client contacts S0
Request: Resolve Root:[edu, memphis, cs]
S0 looks up "edu" in the Root directory table and identifies D1.
Response: A reference to S1, which manages D1.
Step 2: Client contacts S1
Request: Resolve D1:[memphis, cs]
S1 looks up "memphis" in the D1 directory table and identifies D3.
Response: A reference to S2, which manages D3.
Step 3: Client contacts S2
Request: Resolve D3:[cs]
S2 looks up "cs" in the D3 directory table and identifies L1.
Response: 192.168.1.10
Final Result:
Root:[edu, memphis, cs] → 192.168.1.10